11th UNESCO Africa Engineering Week & 9th African Engineering Conference

AI AND IOT-ENABLED PPE COMPLIANCE AND ACCESS MANAGEMENT SYSTEM

PUTSHU LUNGHE Samuel
Departement of Electronics and Communication Engineering,
ISBAT University Kampala, Uganda

1.PROBLEM/CONTEXT

In the telecommunications sector, ensuring strict compliance with safety protocols is essential before workers can access sensitive facilities such as warehouses and equipment hubs. Workers are expected to wear Personal Protective Equipment (PPE) like helmets, reflective jackets, and safety boots, while also presenting valid identification for entry. However, current reliance on manual checks is inefficient, inconsistent, and prone to human error, offering limited real-time oversight for administrators. This gap increases the risks of workplace accidents, delays in operations, and regulatory non-compliance, while also leaving facilities vulnerable to unauthorized entry.

2. GOAL & OBJECTIVES

Goal
The project aims to design and demonstrate an automated access control system that verifies both worker authorization and PPE compliance before granting entry to telecom warehouses. By combining RFID-based identification with AI-powered PPE detection, the system ensures safe, secure, and efficient access while providing administrators with real-time visibility and alerts.

Objectives

- Integrate RFID-based identification with PPE compliance checks.
- Use computer vision (YOLO model) to detect helmets, jackets, and boots.
- Log results into a real-time dashboard for administrators, and the dashboard is also used for new cards registration.
- Send SMS (using the Africa's Talking API) and email (using EmailJS API) notifications for compliance and access events.
- Build a scalable, low-cost prototype adaptable to other industries.

5. CONCLUSION & NEXT STEPS

This project demonstrated that combining RFID authorization with AI-based PPE detection creates a reliable and low-cost solution for enforcing safety compliance. By integrating real-time logging and notifications, the system improves worker safety, operational efficiency, and accountability in telecom warehouses and beyond.

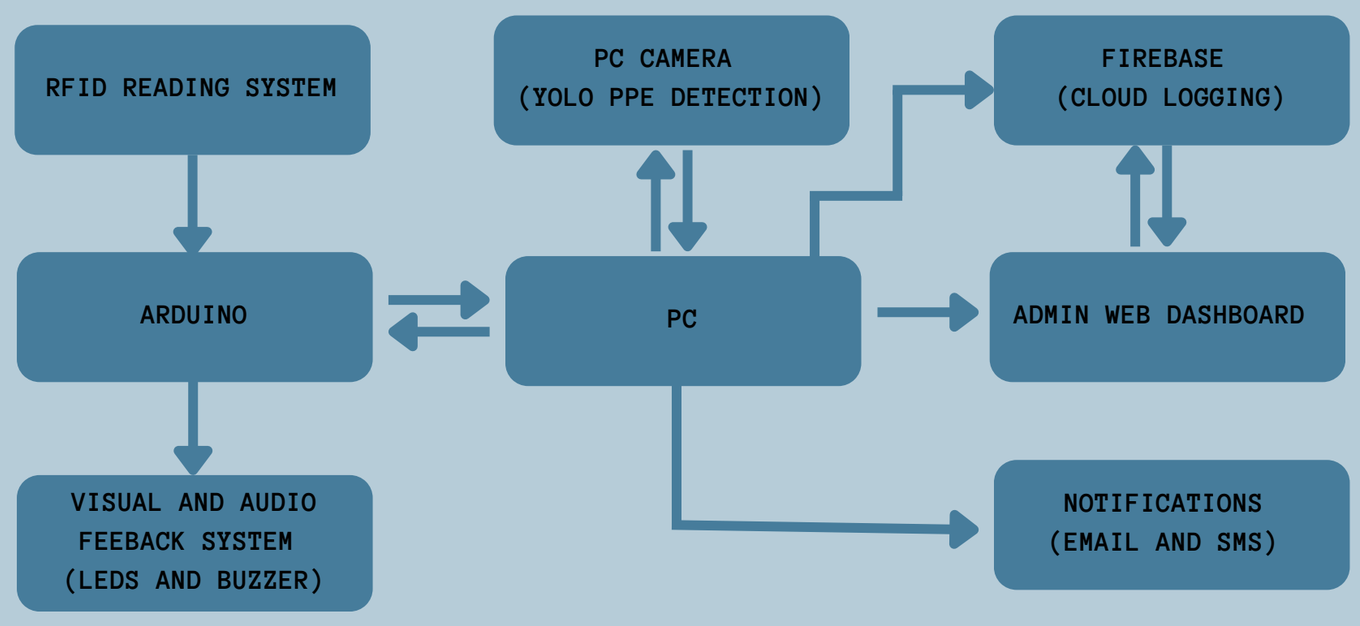
Future development will focus on improving detection accuracy in varied environments, expanding PPE categories or adopting the system to another space, deploying edge devices like Raspberry Pi for portability, and exploring biometric integration for stronger access control. Partnerships with industry players will be key to scaling this solution to real-world deployments.

6. ACKNOWLEDGEMENTS & CONTACT

Acknowledgements
Special thanks to my mentors (Dr. Mayur Kumar Chhipa, Mister Bwenge Kyangwi Josh), colleagues, and peers who guided me throughout the project. I also acknowledge the open-source communities behind YOLO, Firebase, and Arduino for providing the tools that made this prototype possible.

Contact
PUTSHU LUNGHE Samuel
samuellunghe@gmail.com
+256742618584
IEEE Membership Number: 98744354
Finalist, Bachelor of Engineering in Electronics and Communication, ISBAT University

3. SYSTEM ARCHITECTURE & METHODOLOGY



Workflow

1.Card Verification
A worker taps their RFID card on the reader. The Arduino sends the UID to the PC, which checks validity in Firebase. In case it is not valid the process does not go to step 2.

2. PPE Detection
If the card is valid, the camera activates and runs the YOLO model to detect required PPE (it can be helmet, safety vest, jacket, or boots, But in our case we picked safety vest for simplicity).

3. Decision Layer

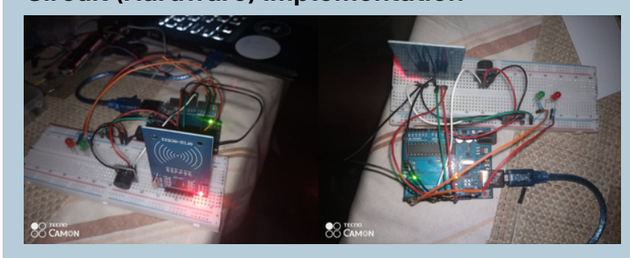
- If PPE is compliant → system sends "PPE_OK" to Arduino → access granted.
- If not compliant → "PPE_FAIL" is sent → access denied, buzzer alert.and requires the user (techinician) to tap the card and restart the whole process again.

4. Data Logging
Results (card ID, PPE status, timestamp, image evidence) are logged into Firebase in a case of a successful PPE Check.

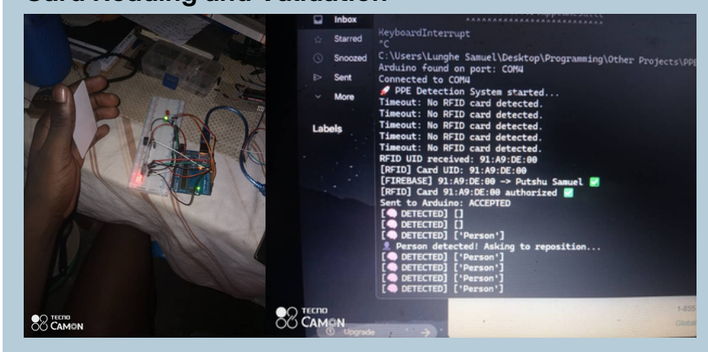
5. Real-Time Monitoring
Supervisors/Admins receive SMS/email notifications and can review entries in the web dashboard instantly.

4. RESULTS & ANALYSIS

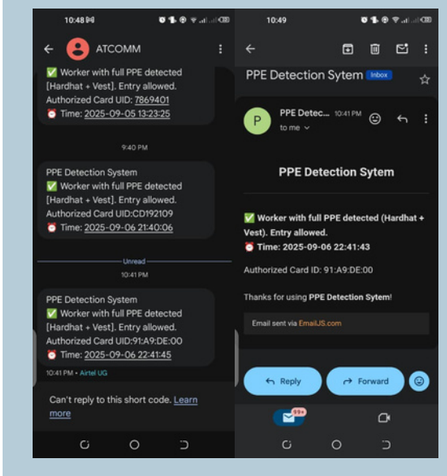
Circuit (Hardware) Implementation



Card Reading and Validation

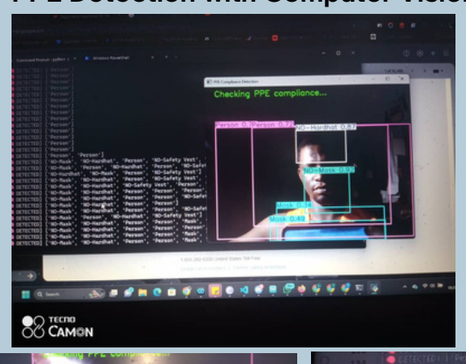


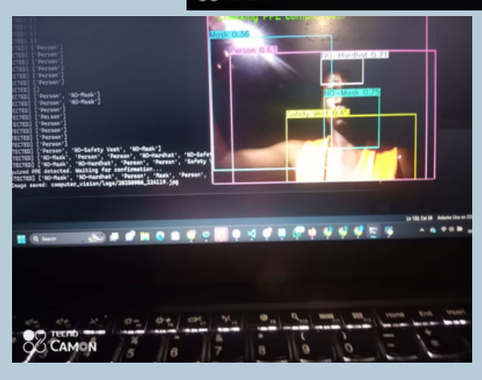
SMS and Email Notifications

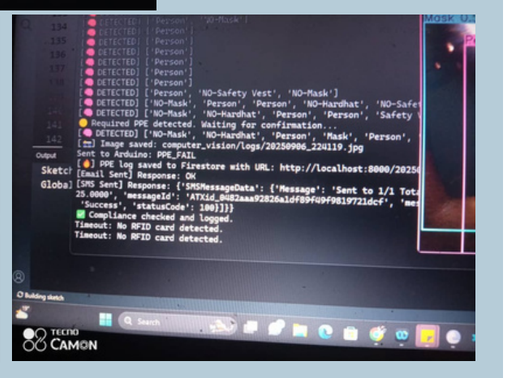
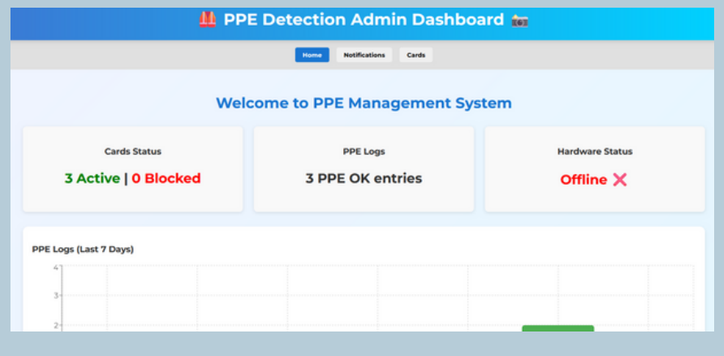
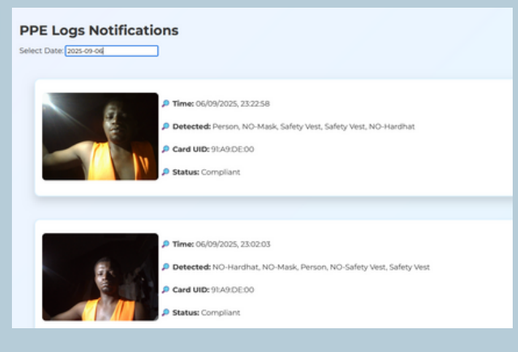


The prototype system successfully combined RFID authorization with PPE detection and real-time logging. Authorized cards reliably triggered the camera check, and detection achieved about 85% accuracy for helmets and vests, and 75% for boots under controlled conditions. Each access attempt was logged in Firebase with card ID, compliance status, and timestamp, while the dashboard updated in real time. Notifications via SMS and email reached supervisors within 5 seconds, ensuring timely oversight. Overall, the system reduced unsafe entries and provided a reliable digital audit trail to improve safety compliance and efficiency.

PPE Detection with Computer Vision (YOLO)







Admin Web Dashboard